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Optimizing smart home energy management for sustainability using machine learning techniques

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Abstract
Energy is fundamental to all significant human endeavors and is crucial for sustaining life and realizing human potential. With the advent of smart homes, energy consumption is increasing as new technologies are introduced, leading to shifts in both lifestyle and societal norms. This scenario presents a unique energy challenge that requires extraordinary efforts to meet the anticipated energy demands. Various innovative strategies are being implemented to overcome the drawbacks and address the growing consumer demand for energy. Today, smart homes offer much more than just basic functions; they also focus on resource management, energy efficiency, and enhancing quality of life. Machine Learning (ML) plays a vital role in smart homes as it allows for the training, adjustment, and optimization of various functions. This intelligent, purposeful capacity has the potential to turn homes into dynamic and practical environments that improve daily performance, ease, and personalization. In this research, an ML-based multivariate model is proposed utilizing Long Short-Term Memory (LSTM) for smart homes, aiming to optimize energy utilization and improve management in the realm of energy consumption. This model offers precise predictions of energy consumption, ensuring minimal random errors. Prominent metrics include a low Mean Squared Error (MSE) of 0.02284, a high Mean Absolute Error (MAE) of 0.184, a Mean Absolute Percentage Error (MAPE) of 0.123, the lowest Root Mean Squared Error (RMSE) at 0.15113, a significant Mean Absolute Scaled Error (MASE) of 0.996, and a strong R-squared value (R^2) of 0.694. The proposed model delivers exceptional predictive performance as compared to the previous approaches, ensuring high reliability, which aligns with the standards needed for advancing toward a smart and sustainable future.

Keywords Sustainable energy management · Smart home technology · LSTM · HEMS · ML · MSE · MAE · MAPE · RMSE · R2

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1 Introduction

The Home Energy Management System (HEMS) has emerged as a key player in bridging the energy market and its digital transformation. HEMS allows customers to track, manage, and regulate their home energy consumption, providing real-time alerts and reports on their usage patterns. The rise of energy consumption within the smart homes poses quite some complexities when used at a very fast pace thus calling for efficient management for the usage by intelligent HEMS. From its inception, this technology has focused on giving users the tools to efficiently record, monitor, and control their home energy consumption. As the market for HEMS continues to expand, new systems have been developed with advanced features, such as demand control, load management, and peak shaving, thanks to ongoing technological advancements. Numerous companies have entered the market, with over fifty firms now offering HEMS solutions across Europe, contributing significantly to the industry's growth [1–3].

A critical element of HEMS is the use of smart energy meters, which, unlike traditional meters, allow for two-way communication. This two-way flow of information occurs between the meters and various strategies and among the meters themselves [4, 5]. The key benefits of this communication are:

- Providing consumers with real-time information on energy utilization and pricing, allows them to adjust their consumption and manage costs effectively.
- Enhancing system security and reducing operational costs for providers through remote functions like diagnostics, validation, and load monitoring.
- Improving overall energy efficiency by balancing production and consumption, minimizing excess energy generation, and reducing greenhouse gas emissions.

The integration of these mechanisms forms the organization of HEMS. The system's operations are overseen by a software platform that uses Information and Communication Technology (ICT) to coordinate between the core HEMS unit, sensors, and smart devices within the home. ICT plays a central role in data transmission, particularly in setups where intelligent load scheduling and pattern recognition are essential [6].

Artificial intelligence (AI) is heavily integrated into HEMS, enabling automatic decisions and responses based on data collected by sensors. For instance, techniques like peak shaving are used to optimize energy usage by avoiding high-demand periods, thus lowering costs for consumers or reducing strain on the grid [7]. Additionally, solar photovoltaic (PV) systems and grid-connected storage options help boost energy self-consumption, reduce costs, and ease grid pressure. Smart home appliances, like refrigerators, stoves, and TVs, are equipped with software that allows them to communicate with the HEMS. Even traditional appliances can be made "smart" by using smart plugs, enabling remote control and monitoring. The HEMS system provides feedback on appliance energy consumption, helping users understand and optimize their energy use based on their household behavior [8–10].

ML is becoming a vital aspect of HEMS, enabling systems to autonomously learn from data and make informed decisions deprived of human intervention. ML's ability to analyze large datasets and predict outcomes helps in improving energy management, making HEMS smarter and more efficient over time. ML, especially LSTM models, plays a pivotal role in enhancing energy predictions and reducing errors in these systems. In this study, specific factors stimulating the need to develop efficient sustainable energy solutions pu into consideration the prospect of enhancing the energy prediction models in smart homes.

2 Literature review

Many studies have highlighted the significance of thermostatic regulators and the physical management of energy loads within HEMS. Devices such as heaters, air conditioning units, also heat pumps, particularly HVAC systems, are major suppliers to energy consumption during peak periods. Deliberate scheduling and load shedding for these devices are promising techniques to reduce overall energy use, as explored in various studies on load management for domestic utilization.

In [11, 12], methods like Binary Particle Swarm Optimization (BPSO) and fuzzy Mamdani systems were designed to monitor and schedule electrical loads across ten single-family homes. BPSO optimized appliance usage during non-peak times, while fuzzy logic managed irregular-use devices such as air conditioners. While the thermostat settings

are intended to improve energy efficiency, they sometimes compromise user ease. Literature has also recommended a fuzzy regulator for passive cooling in residential buildings.

In this context, the revolutionary potential of IoT and distributed computing was also studied in the smart home industry. In creating the basics of the future network-based home services, these technologies are essential including the possibility of developing smart city infrastructures [13–15]. A study presented a novel DT approach to keep the clean and precise energy forecast models with the help of misclassified output while filtering noises. In the preprocessing phase, the Misclassified Recall (MR) technique was employed to correct the misclassified cases especially due to data transmission errors or faulty devices in energy prediction. Simulations on datasets forecasting energy usage in low-energy buildings showed that this approach achieved better accuracy than existing methods [16].

Renewable energy integration and energy efficiency are essential for long-term energy transitions and mitigating climate change. This study examined the significant role of the Internet of Things (IoT) and modern technologies in various aspects of the energy sector, including trade, manufacturing, communication, circulation, and consumption. The study emphasized IoT's probable to improve energy efficiency, indorse the use of renewable energy, and reduce environmental impact, while also addressing security in IoT integration and suggesting solutions like blockchain technology. These insights were valuable for energy legislators and administrators aiming to optimize the energy sector [17].

In another study, the emphasis was made on the progressive significance of energy saving and comfort in residential facilities to reduce energy consumption. Specifically, the study focused on heating systems that used strategies that control the users' comfort as well as the flow of energy. Different control strategies were applied to tweak values and minimize energy usage while ensuring proper climates. Energy clinics carried out for the smart homes in the study enhanced thermal, visual and air quality comfort aspects on which the research focused [18]. In another study, a mixed-integer linear programming system was developed for scheduling household appliances. This method aimed to lower electricity costs by shifting loads to off-peak hours, ultimately reducing energy consumption, lowering peak power usage, and efficiently managing smart appliances controlled through power signals [19].

A novel model for managing electricity loads in smart homes was proposed in [20]. The model incorporated load definition, backup mechanisms, and a clear interface, allowing users to classify and control loads efficiently. In a related study, a dynamic estimating system using the diverse knapsack technique placed consumer appliances on a variable peak pricing schedule, enabling households to achieve cost savings on electricity [21]. Techniques like co-evolutionary particle swarm optimization were also introduced for scheduling loads to maximize benefits for homeowners [22].

Another study presented a home switch system equipped with machine sensors collecting energy data from all of the household appliances. It interoperates with a public broker server, enabling it to establish connections with home network devices such as security cameras. The work also tried to establish between Message Queuing Telemetry Transport Protocol (MQTTP) and Hypertext Transfer Protocol (HTTP) protocols to know which one of the methods is suitable for controlling home services even though the design did not comprise Big Data for managing large data obtained from different home sensors [23].

In [24], researchers discussed the development of a DC distribution system for homes centered on DC-powered devices connected through IoT. However, the widespread adoption of smart DC-powered homes faced obstacles due to the lack of common standards and protocols. Integrating IoT into such systems could potentially address these challenges.

In [25], the study explored the use of In-Home Display Systems (IHDs) and Automatic Meter Reading (AMR) for energy management communication. The research showed that smart home systems could select display methods such as TVs, smartphones, or tablets based on the environment and user preference. However, concerns arose about the need for a standardized user interface across all home systems to meet the diverse needs of users.

In [26], a proposed HEMS architecture used power line communication to provide real-time monitoring of home energy consumption. The system allowed users to control appliances remotely through smart meter data, although it relied on the traditional HTTP protocol rather than lighter options like MQTTP, which may be essential for expanding the system to cover more households. In [27], a home gateway manager was implemented with an administration system that created active schedules based on weather conditions for all connected knocks in a home net. The web server retrieved device statuses as well as energy utilization data over an XML border, but the plan faced challenges when transferring large XML files across the network, which could pose capacity issues when handling large data transfers between browsers and servers.

The authors in [28] proposed using programmed algorithms as a method for controlling lighting systems. They used a neural network model to show that this method can guarantee overall and specific savings in the energy of residential lighting systems. To forecast the efficiency of energy conservation, the obtained results of the input vectors were

represented in the neural network. Moreover, another study [29] brought into use a household robotics system for home energy management along with the prediction layer that delivered energy as per the expected household events.

In a study, the authors transformed a load scheduling problem into a load commitment problem [30]. They redefined this complex problem as a multi-stage decision-making process, akin to a Markov decision problem, and developed a solution using reinforcement learning techniques to address it effectively.

A framework for managing energy consumption in multi-resident smart homes was also introduced in another study [31]. This system leveraged mobility-aware resources and applied game theory to minimize uncertainties by optimizing utility functions. The research team further created a demand-side management simulation tool that incorporated dynamic resource allocation methods [32, 33]. This hybrid energy management system was used in simulations with household appliances to implement resource management strategies effectively.

In [34], the authors examined methods for energy conservation through data-driven feedback, with a specific focus on electronic feedback provided by smart meters. They emphasized the significance of collecting feedback at regular intervals to improve the design and evaluation of energy consumption meters [35]. Another study [36] proposed a smart energy management method that balanced consumer needs with energy conservation, based on advanced user preferences and automatic device switches [36, 37]. Some of the new approaches that have been considered in energy management are as follows; [38] proposed a fuzzy-based federated learning model for sustainable energy management with integration based on IoE. The study [39] used predictive analytics for improving the electricity consumption resilience. Also [40] employed MapReduce and private blockchain for smart city traffic prediction that could be potentially helpful for efficient use of energy. Kaluri et al. proposed a rough set theory-based model with a Deep Neural Network for accurate battery life prediction in IoT devices. Similarly, recent work on HEMS used machine learning to optimize energy consumption in smart homes [43, 44].

Table 1 offers a detailed overview of various technological methods aimed at optimizing energy usage in residential settings. The various studies reveal different research findings and solutions for scheduling the appliances using algorithms such as BPSO and DT or using IoT accompanied by cloud computing for energy trading. The approaches presented include the use of machine sensors and mixed-integer linear programming to increase energy efficiency. Nonetheless, challenges such as security issues, limitations in data management, and the effectiveness of protocols reveal the complexities of implementing smart energy systems in residential environments.

3 Methodology

In recent times, the growing emphasis on improving energy efficiency in residential buildings has highlighted significant challenges in optimizing energy use. Tackling this issue is crucial for achieving better energy management in the sector. ML has introduced a game-changing approach, reshaping the way smart homes function with its advanced algorithms that enable intelligent adjustments, enhance operational efficiency, and optimize daily activities. This research proposed a dynamic energy management model that addresses the demand for sustainable energy solutions, paving the way for smarter energy consumption in homes. The proposed model, as illustrated in Fig. 1, leverages ML to transform how energy is utilized in smart home environments.

Figure 1 presents the structure of the proposed Energy Consumption Model (ECM), which aligns with recent progressions in communication technologies. The model is composed of several layers, including an input, a preprocessing, a training, and a validation phase. The ECM allows energy devices to interconnect efficiently, enabling the exchange of energy data. This setup enhances connectivity by collecting sensory data as of wirelessly coupled devices, avoiding the requirement for expensive cellular networks. With the ECM, data can be gathered from different energy sources also transferred to the next layer for real-time monitoring as well as efficient energy management. The sensed data are processed through the preprocessing, training, and performance layers. Subsequently passing through the performance stage, the effectiveness moves to cloud computing. In validation, the learned or trained models are imported from the cloud to forecast and assess energy management actions.

Figure 2 illustrates the proposed model data flow diagram, divided into two main phases: training as well as validation. The training phase is comprised of four layers. Firstly, energy sensors installed at various locations gather the data. This dataset is then moved to the essential preprocessing layer of the proposed ECS, where techniques such as normalization as well as moving averages are executed to handle missing values also bring all values within a precise range. The preprocessed data is then split into 70% (training) also 30% (validation) to improve the classification process for better accuracy. Subsequently, the training data is forwarded to the training layer while the validation data is archived in the

Table 1 Comprehensive analysis of the literature review

References	Technique	Concentration	Outcome
[11, 12]	BPSO Fuzzy Sugeno	Detecting and Scheduling Electrical Loads	Methods executed in ten single-family apartments, tackling energy consumption scheduling and unequal device use
[16]	Advanced DT, Misclassified Recall	Filtering Noisy Data for Prediction Models	Simulation experiments demonstrate superior accuracy
[17]	IoT, Cloud Computing, Blockchain	Impactful Contributions in Energy Trade, Production, etc	Valued perceptions for energy policymakers, economic experts, and directors seeking to optimize the energy sector through IoT integration
[19]	Mixed-Integer Linear Programming	Household Appliance Scheduling for Cost Reduction	Condensed electricity charges, and amended operational efficiency
[23]	Machine Sensors, MQTT, HTTP	Home Switch System for Consolidated Energy Data Delivery	consolidated energy data delivery, associating MQTT and HTTP for efficiency
[27]	Home Gateway Organizer, XML	Operational Schedule Based on Meteorological Situations	Probable volume when transferring large XML files in the network

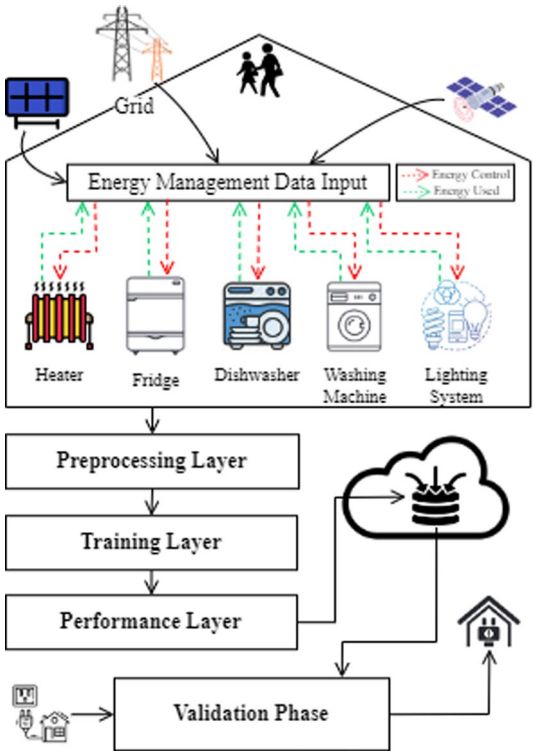


Fig. 1 Proposed model abstraction

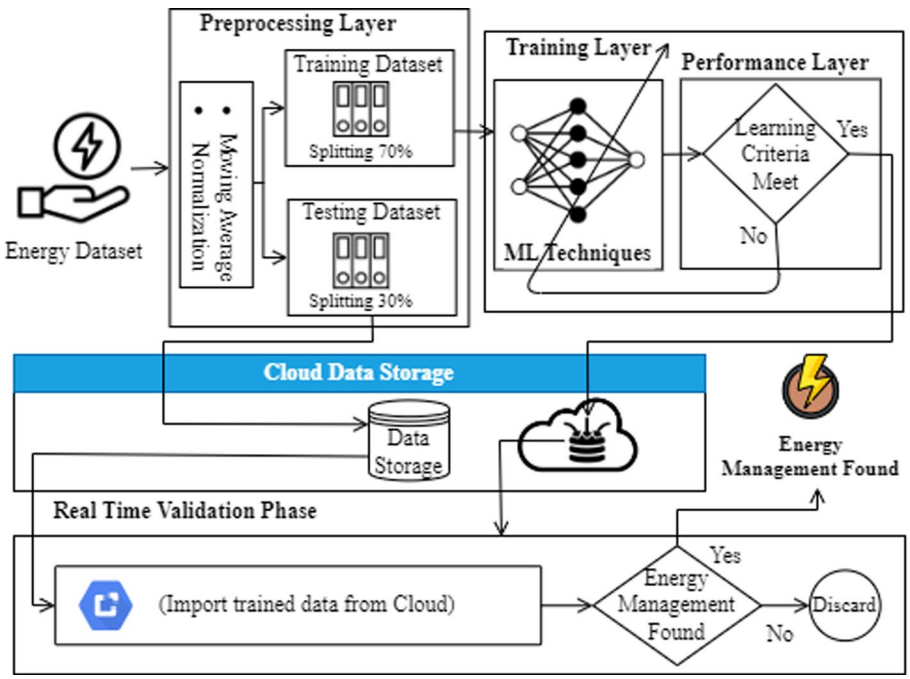


Fig. 2 Proposed model data flow diagram

cloud. This training layer then uses an ML method to categorize and detect energy usage. The output is then taken to the performance layer where the energy used is estimated by the levels of accuracy or the error rates. According to the learning criteria, if the required standards have not been attained, there will be a change in the training layer. However, in the case the criteria are met, the output generated from the ML technique is stored in the cloud data storage.

The remaining 30% of testing data is imported from the cloud in the validation phase and the learned model from the cloud data storage for the prediction of whether the energy consumption is found or not. If the answer is 'No' the process is terminated; in the case, if the answer is 'Yes', the message is shown as 'Energy management is found'.

3.1 Dataset description

Table 2 gives the 1-minute interval reading recorded at a stipulated period for 350 days. It measures and records the amount of electricity used in kilowatts (kW) by different appliances in a household according to smart meters. Because of this, the large data set provides important information regarding the energy consumption of household appliances over several years.

Figure 3 shows the correlation matrix of various energy-related attributes in the form of a heatmap. The heatmap is constructed to include two variables on the horizontal and vertical axis of the grid and the correlation coefficients next to each grid square. These coefficients are from − 1 to 1, where − 1 represents a strong inverse correlation, 1 represents a strong positive correlation and 0 indicates no relation at all. This visual representation helps to clarify how different energy variables are interconnected over time. High correlations can indicate either redundancy or dependence between features, which could potentially complicate data analysis and modeling efforts.

Figure 4 depicts two spatial plots in a single figure but vertically placed one upon another. The first plot on top shows the mean energy requirements for both 'use' and 'House overall' in two indices 'Daily' and 'Average' (values per day). The bottom plot for 'gen' and 'Solar' shows the oscillation of the daily mean and how it frequently changes as seen in the first plot where the data is aggregated to the daily intervals and average. The second plot provides one overview of the daily energy consumption and production, pointing to the fact that energy use in households has become an essential part of daily life.

In Fig. 5, a heatmap shows weather-related attributes about one another. These collation plots illustrate pairwise associations and conjunctions of the chosen weather attributes, where the heat mapping helps in distinguishing positive from negative associations and their magnitudes. Dark blue is used to denote a negative correlation coefficient while dark red is used for positive correlation coefficient and white for no correlation coefficient. The heatmap is made up of squares each of which has a numerical value that correlates to the correlation coefficients. This helps in the quick and easy understanding of the relationships between numerous weather attributes in the dataset.

Figure 6 shows the temporal distribution of meteorological parameters in two parallel plots where daily mean values of important factors are shown. The first subplot shows time-averaged values of 'temperature', 'apparent temperature', and dew Point, which are samples of these quantities over time. The second subplot on the bottom provides information about the daily mean values of 'humidity', giving an idea of the tendencies in the concentration of moisture.

Figure 7 presents the dual-subplot visualization of the meteorological parameters where the temporal development is shown. The first subplot located at the top is the brief visualization of the average daily fluctuation of 'Tdiff' and 'windSpeed' where their raw data is sampled at a daily frequency to estimate their daily mean values. The bottom subplot is all about making a direct comparison between the two variables, namely, 'apparent temperature', and 'temperature', but focusing on the temporal characteristics of the data. This figure is useful to illustrate the time variation in the temperature-related aspects and wind speed and thereby offers a better view of the main meteorological parameters.

Figures 8 and 9 illustrate the initial six columns of the dataset vary over time. Each subplot in these figures represents a different feature and shows the daily average values, which have been adjusted to reflect daily means. The features under consideration can be regarded as different faces of the dataset, for instance, different variables or measurements.

Table 2 Dataset attributes description [45]

Features	Description
Time	Time
use [kW]	Total energy consumption
gen [kW]	Total energy generated through solar or other power generation resources
House overall [kW]	Overall house energy consumption
Dishwasher [kW]	Energy consumed by a specific appliance
Furnace 1 [kW]	Energy consumed by a specific appliance
Furnace 2 [kW]	Energy consumed by a specific appliance
Home office [kW]	Energy consumed by a specific appliance
Fridge [kW]	Energy consumed by a specific appliance
Wine cellar [kW]	Energy consumed by a specific appliance

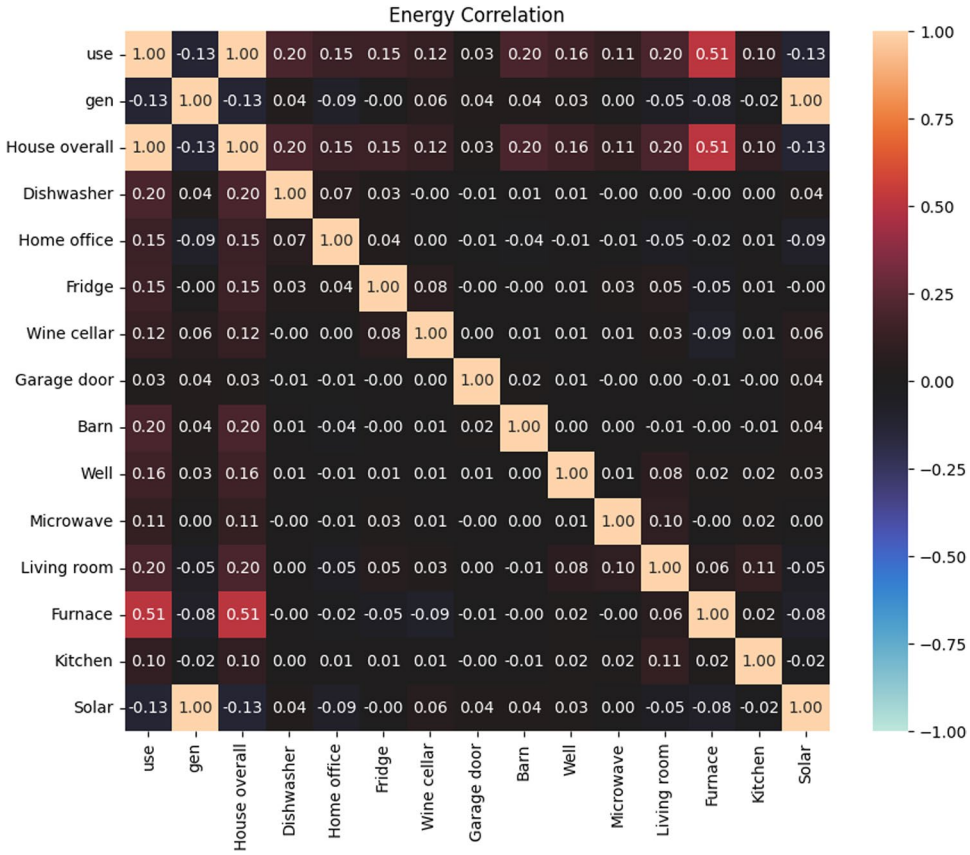


Fig. 3 Correlation matrix of energy attributes

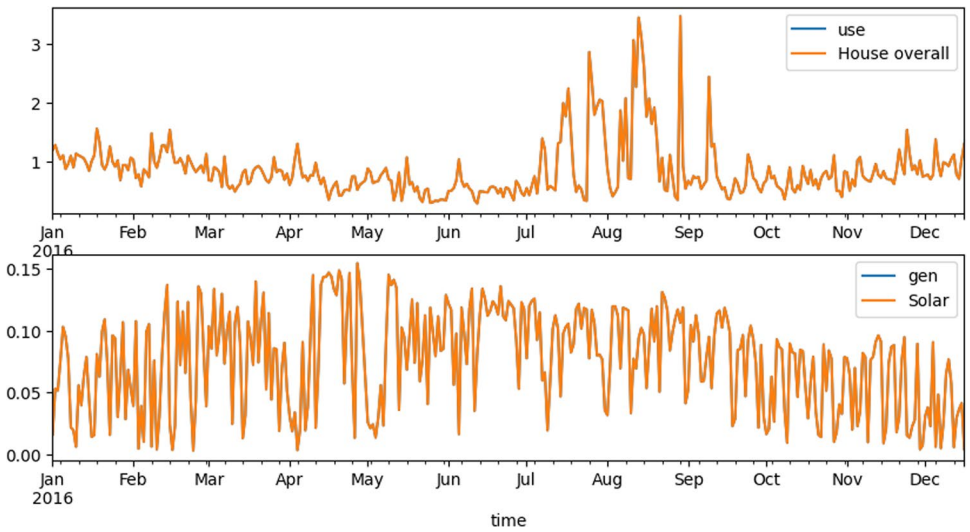


Fig. 4 Generation patterns and daily energy consumption

The grids also help illustrate temporal patterns/variation across the chosen features simultaneously for convenient bundled evaluation.

In Fig. 10, the overall comparison of the dataset is shown in terms of mean values by month. The first subplot in each of the layouts that form the grid displays the individual monthly means of the features mentioned above. The grid is presented in a row with three columns enabling the comparison between the months side by side. To enhance clarity, markers ('o') are used to highlight key data points, making it easier to spot significant patterns. This visualization is useful for identifying both short-term fluctuations and long-term dynamics in the dataset, as well as observing seasonal variations across the selected features.

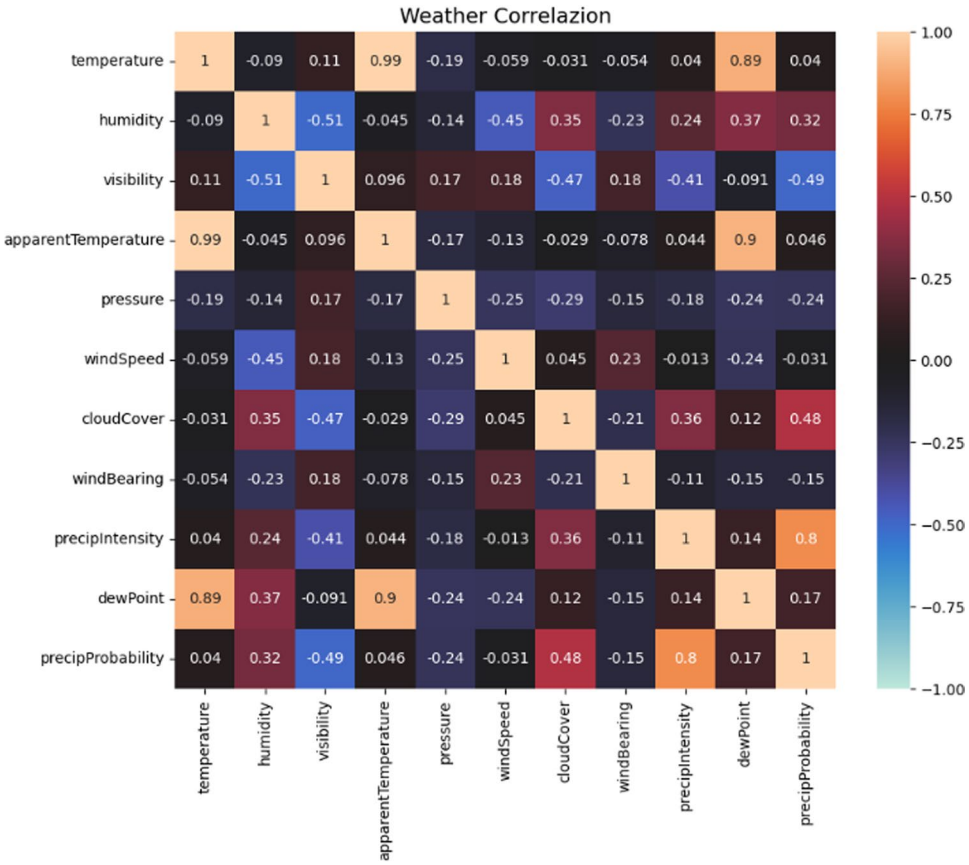


Fig. 5 Correlation heatmap of weather

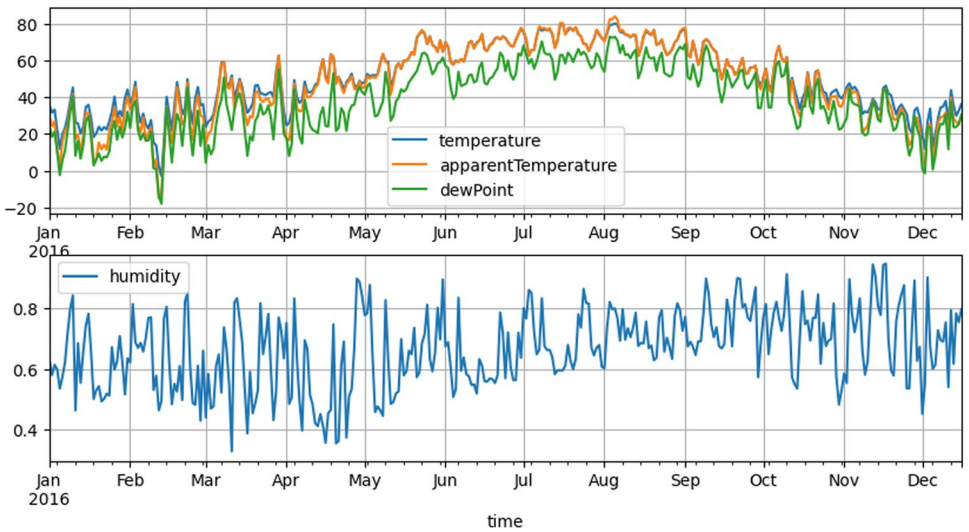


Fig. 6 Meteorological parameters analysis in climate data

Figure 11 presents the distribution in mean values of the attributes of the dataset that have been accumulated based on the days of the week. Markers ('o') are applied to make the data points more visible, while the grid enhances readability, making it easier to identify weekly patterns. This figure allows for observing the differences and trends in the selected features, offering insights into how the dataset changes for a week through regular analysis.

Figure 12 represents the average consumption per hour of the selected dataset attributes. Separately subplot focuses on a specific feature, displaying its mean values across different hours of the day. The grid layout allows for an easy comparison of hourly trends between the features. Dot plots ('o') highlight key data points, while the grid aids in interpreting the hourly data more effectively. This visualization reveals peak consumption times and recurring patterns within

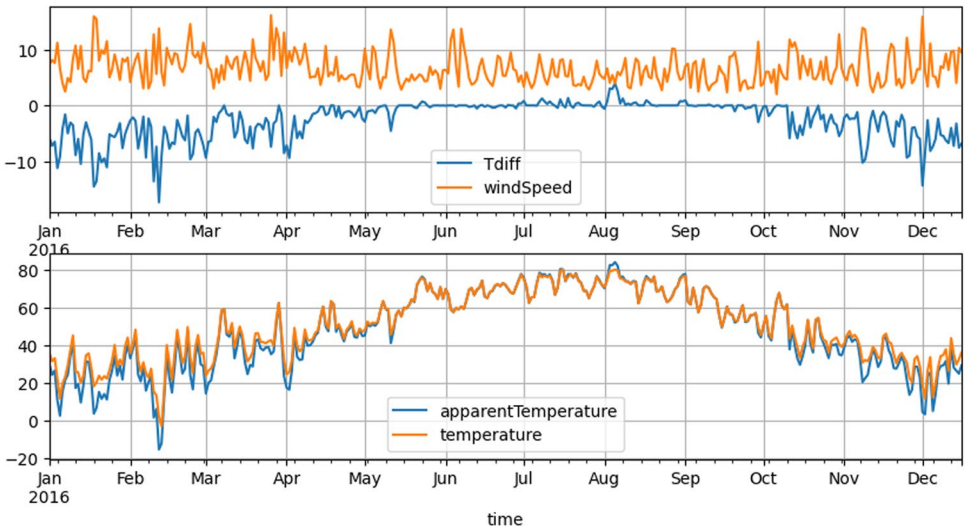


Fig. 7 Meteorological parameters temporal analysis

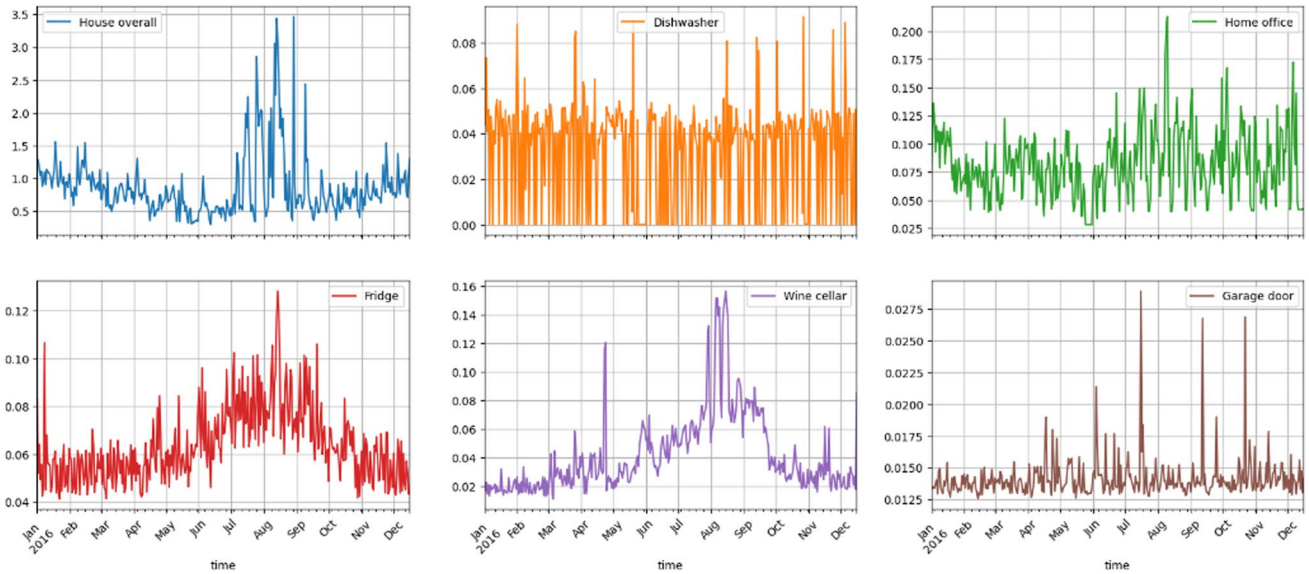


Fig. 8 Temporal evaluation of resampled daily mean data

the dataset, helping to identify the hours when usage is at it's highest. It offers valuable insights into how consumption behavior changes throughout the day and provides clarity on peak usage times.

4 Simulation results

This study underscores the importance of addressing challenges in accurate prediction. After thoroughly evaluating several models, including Baseline, Auto-Regressive Integrated Moving Average (ARIMA), SARIMAX, and LSTM, the research concluded that the LSTM Multivariate model demonstrated the strongest predictive ability among them. This proposed study utilized various visualizations, such as Seaborn bar charts, grouped bar charts, and scatter plots, to highlight the superior performance of the LSTM Multivariate model. It focuses on exploring the challenges and solutions within the decision support model, with the LSTM Multivariate model serving as a key example.

Figure 13 presents the daily energy fluctuations for the 'House overall' feature as well as the 10-day rolling average with an added baseline model. This figure is meant to show trends, variations, and even trends in the use of energy over some time. By applying the rolling mean, short-term variations are minimized, allowing a clearer focus on ongoing changes and making it easier to identify underlying consumption patterns. This visualization provides valuable insights into the

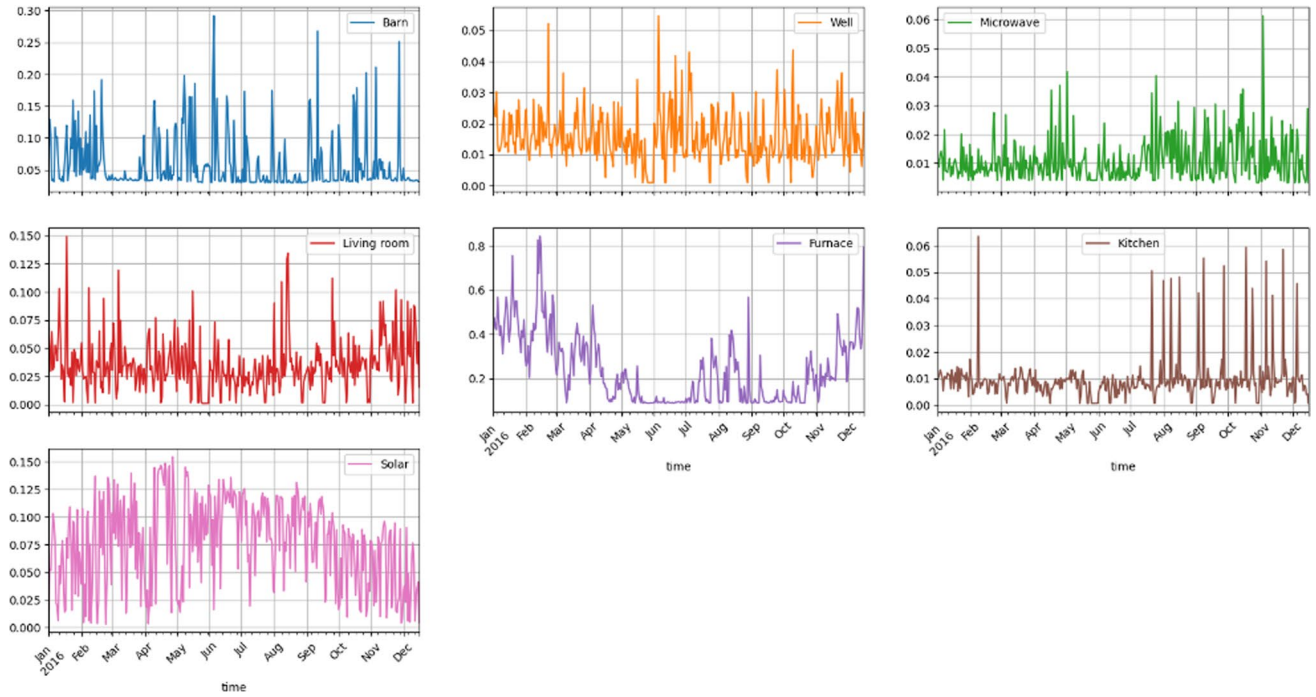


Fig. 9 Temporal evaluation of resampled daily mean data

relationships between energy usage, the reasons for emerging trends, as well as the choice of their positive or negative progression over the practical period.

Figure 14 illustrates the daily energy consumption pattern (depicted in blue) alongside predictions generated by a baseline model (shown in red) in a typical training model. This model uses data from previous days to forecast daily consumption. By comparing these visual representations, the proposed model can assess whether the ML algorithm effectively captures the existing trends in the training data. The overlap of the two lines not only highlights the model's ability to apply learned patterns but also its effectiveness in adapting to real-world situations.

Figure 15 compares daily energy consumption (represented in blue) with forecasts from the baseline model (shown in red). This model predicts today's consumption based on data from the previous day. The close match between the two lines demonstrates the performance of the algorithm in mirroring observed trends. This visualization helps evaluate the model's effectiveness during the testing phase.

The analysis of the forecasts is given in Figure 16 by using only the 'House overall' daily energy patterns with a baseline model of a 10-day rolling mean. The metrics include an MSE of 0.092, which measures the average size of prediction errors. The RMSE of 0.304 implies the average error magnitude and can be regarded as a moderate fit to the real data. The MAE of 0.194 shows the average sizes of forecasting errors. The MAPE of 0.233% gives a relative performance measure as a percentage of actual values. The MASE of 0.790 compares the prediction performance to a baseline model. The R^2 score of -0.203 shows limitations in the model's ability to explain data variability, highlighting the need for improved forecasting methodologies.

These metrics assess forecasting performance, with lower values indicating better performance. They offer quantitative insights into the method's ability to capture general patterns in daily energy consumption and highlight potential improvements compared to other forecasting methods.

Figure 17 examines the temporal relationships in daily energy consumption. The top-left graph, the Autocorrelation Function (ACF), illustrates how current energy usage relates to past values, revealing potential cycles. In contrast, the top-right subplot shows the Partial Autocorrelation Function (PACF), which provides a detailed view of the impact of specific lag values on time lags, helping to identify significant time lags. Differencing is crucial for addressing data non-stationarity, and ACF and PACF plots help determine when the data is sufficiently stationary for time series modelling. The gray dotted lines indicate the 95% confidence intervals, which are used to assess the significance of correlations.

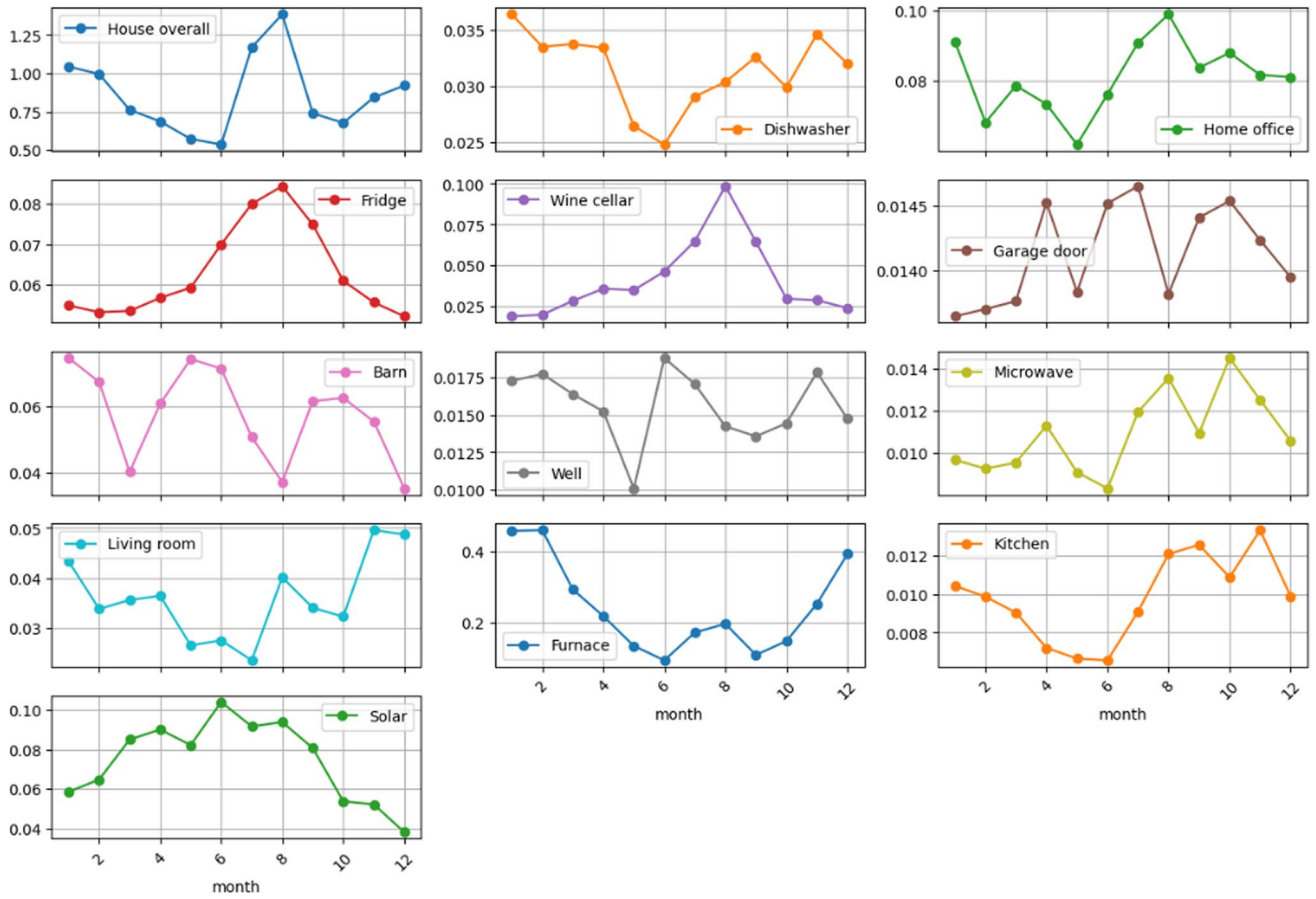


Fig. 10 Monthly trends of mean values for selected dataset attributes

Figure 18 demonstrates the effective performance of an ARIMA model in predicting energy use. The three-part plot is arranged from lowest to top, starting with the Training Data (shown as a greenish line), which highlights the complexity of the past dataset utilized to develop the model. Moving to the Test Data (represented by a bluish line), this phase evaluates how well the model forecasts future energy consumption. The ARIMA model extends beyond historical data to make future predictions. The red and blue lines indicate whether the model is generalizing well or focusing on specific issues.

The shaded Confidence Interval around the red line indicates uncertainty in the model's predictions. A wider interval signifies greater uncertainty, whereas a narrower range suggests higher confidence in the predictions. This diagram illustrates the ARIMA model's learning, testing, and forecasting processes, offering a comprehensive view of its predictive capabilities and the confidence level of its forecasts.

Figure 19 presents key metrics evaluating the ARIMA Time-Series Forecasting model's performance on the test set. The MSE of 0.069 and the RMSE of 0.263 are significant indicators of the model's performance. The MAPE with a value of 0.226, and the MASE with 0.71, suggest the model is performing well compared to the baseline. The R^2 score of 0.095 shows the extent to which the model accounts for the variability in the data. Together, these metrics offer a detailed interpretation of the ARIMA model's prediction performance as well as precision on the test set.

Figure 20 displays the Auto ARIMA regression model applied to predict energy use. The greenish line, beginning from the lower part of the graph, represents the historical test data, highlighting the trends and patterns the model has assimilated over the period. To evaluate the model's predictive performance, the blue line represents the actual test data, allowing a comparison between the model's learned patterns and real-world energy usage. The blue asterisks present the actual observations or the real values of the test set while the red solid line represents the forecasted values obtained from the auto ARIMA model. The shaded area around these predictions symbolizes the 95% confidence interval, which reveals the degree of error. The wider the interval, the higher the number of unknowns involved in the actual or estimated outcomes, the lower the interval, the higher the amount of certainty in those estimations. This figure gives all the details

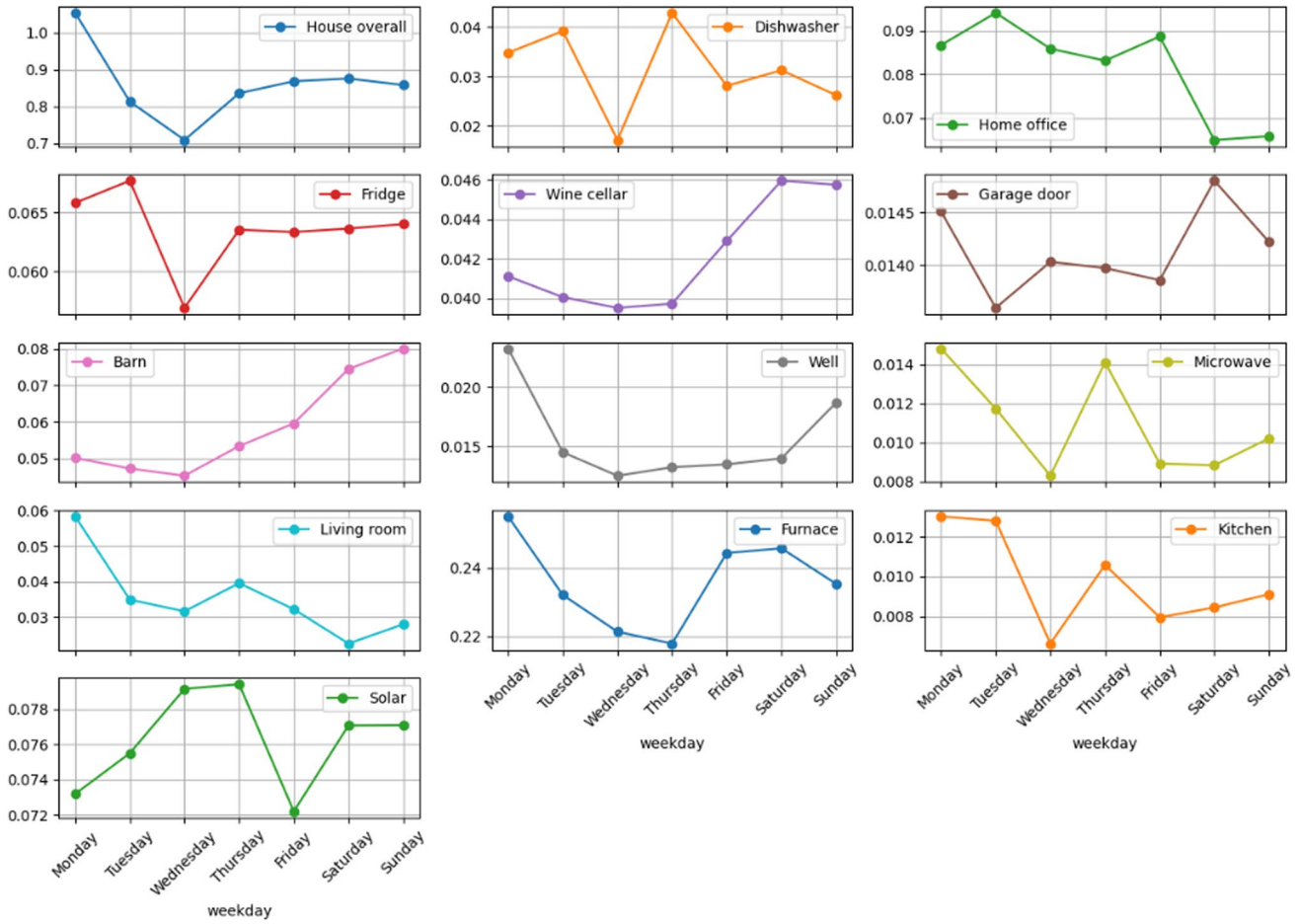


Fig. 11 Trends in selected dataset attributes every week (Mean Values over Time)

of the performance of the auto ARIMA model in its forecasting as well as comparing the actual and the forecasted values besides understanding the level of confidence for the predictions.

Figure 21 provides the evaluation results of time series prediction with the auto ARIMA model for the test data. The MSE of 0.3327 reflects the average squared error between actual and forecast values. The RMSE of 0.581 indicates the precision of the model's predictions. The MAE is 0.469, representing the average variance among forecast as well as actual values. The MAPE of 0.712 provides a percentage-based measure of prediction performance. The MASE of 1.909 compares the model's performance to a naive prediction. The R^2 score of -3.404 highlights the quantity of variance not explained by the model. Together, these parameters offer a thorough view of the model's prediction performance also reliability.

Figure 22 depicts the prediction results of a periodic ARIMA with Exogenous Variables (SARIMAX). The green line states the training while the blue line denotes to the test data. The blue line shows the empirical distribution of the residuals of the test set, while the red line indicates the model's predicted standards for the test sample; the shaded area above and below the red line represents the 95% confidence interval for the forecasted values. This figure practically displays the model on the training and testing stages and the inclusion of external variables to enhance the generalization of the results. The confidence interval also helps to augment the understanding of the fluctuation of the model's predictions.

Figure 23 illustrates the performance metrics for the SARIMAX model evaluated on the test dataset. The MSE is 0.102, reflecting the average squared modification between predicted and actual values. The RMSE is 0.319, which represents the typical magnitude of prediction errors. The MAE stands at 0.245, indicating the average magnitude of absolute errors. The MAPE is 0.366, expressed as a percentage. The MASE is 0.996, suggesting that the model's performance is comparable to a basic benchmark. Nevertheless, the R^2 value of – 0.327 designates that the model does not effectively capture the variability in the test data.

Figure 24 displays the performance of an LSTM Univariate model in predicting future energy use. The blue line represents the training data, showing the learned patterns of the model. The dark blue line shows the validation set used to

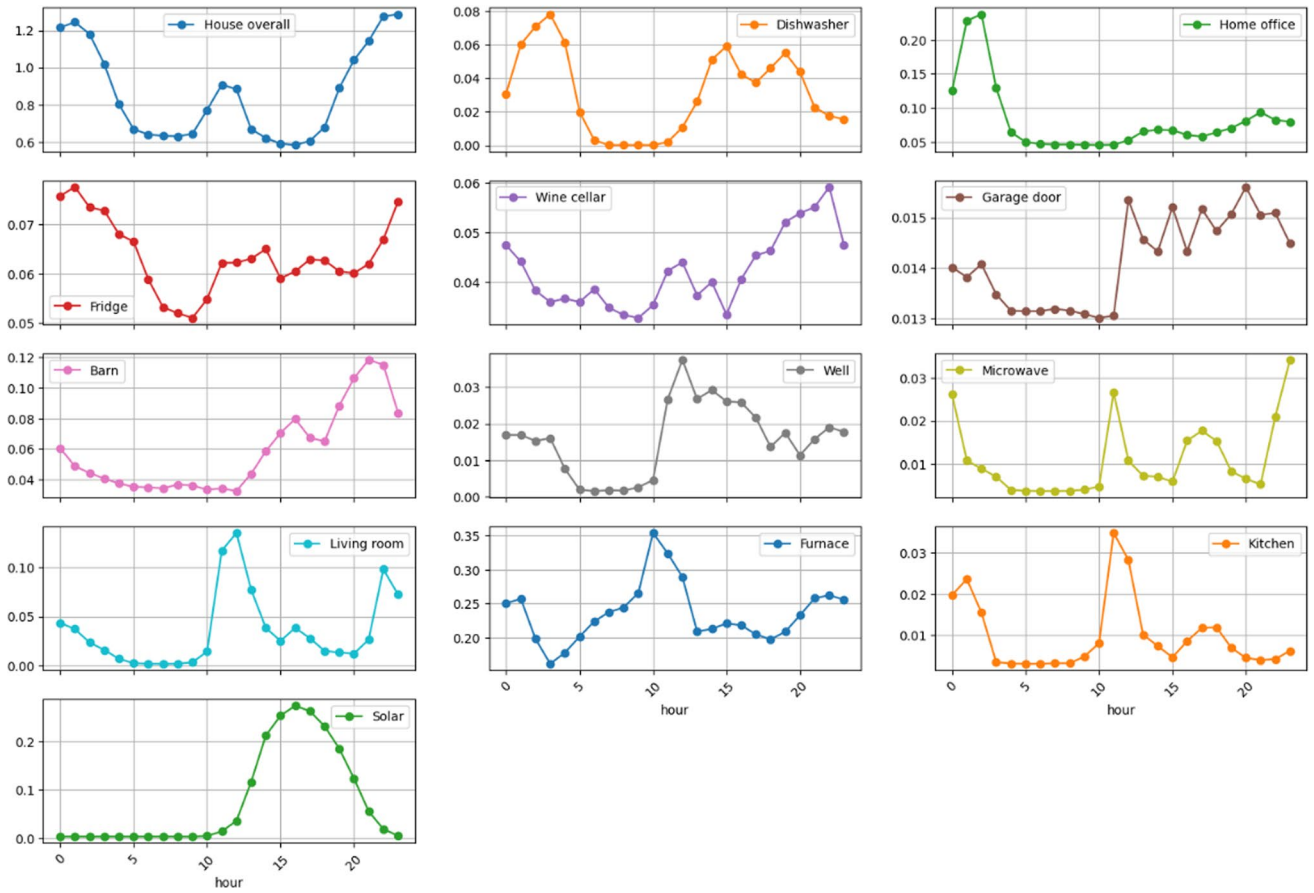


Fig. 12 Hourly trends of mean values for selected dataset attributes

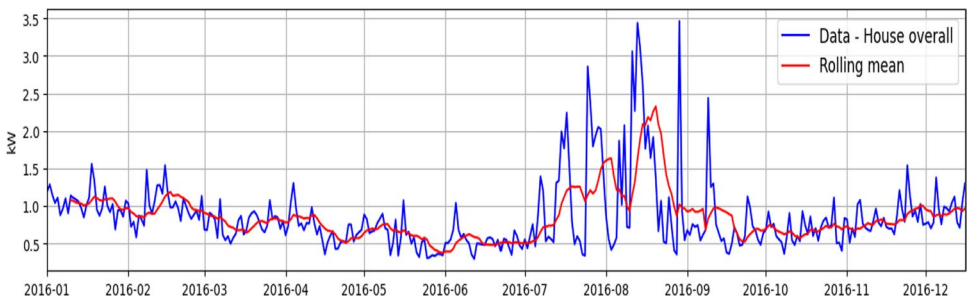


Fig. 13 Daily energy use and 10-day moving mean for 'House Overall'

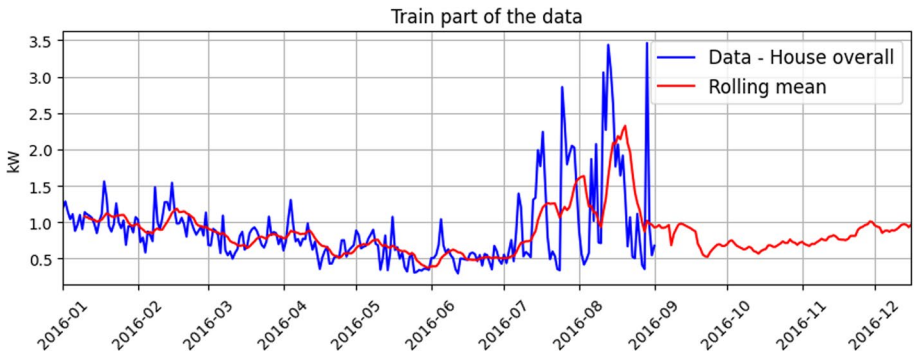


Fig. 14 Baseline model forecast of daily energy consumption (Training)

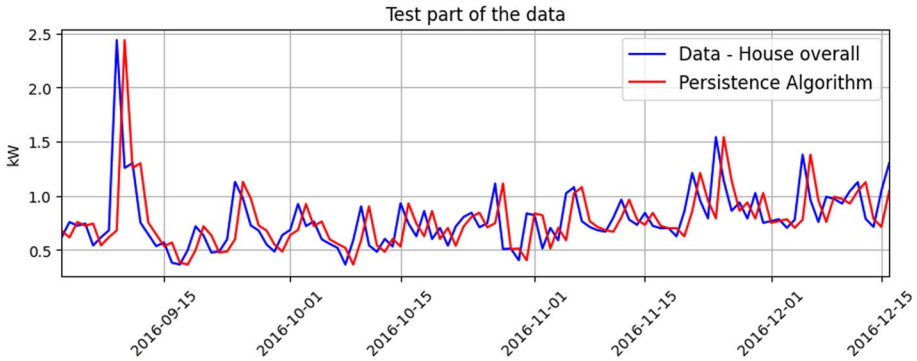


Fig. 15 Baseline model forecast of daily energy consumption (Testing)

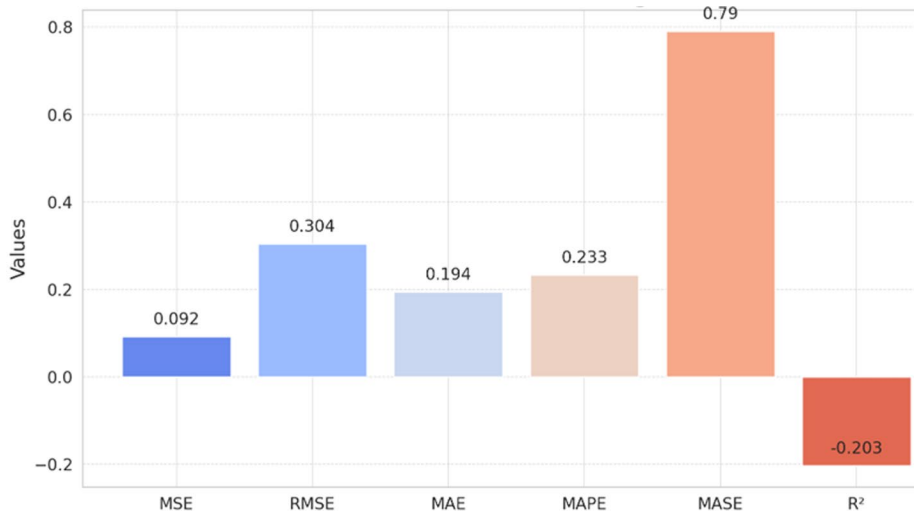


Fig. 16 Baseline model forecasting performance metrics

test the model's prognostic aptitudes. The red line depicts the simulated values for the test data. This visual representation compares actual and predicted values, providing insight into the model's ability to replicate data patterns in the time series. The legend as well as grid assist in evaluating the model's performance.

Figure 25 outlines the performance of the LSTM Univariate model on the test set using key parameters. The MSE is 0.069, representing the average squared error among the foreseen and real values. The RMSE, at 0.263, highlights the average size of these errors. The MAE stands at 0.172, indicating the average complete alteration among predictions and actual outcomes. The MAPE, measuring performance as a percentage, is 0.306, while the MASE value of 0.996 provides a normalized assessment of the model's precision. Additionally, the R^2 score of 0.113 indicates the proportion of variance in the dependent variable explained by the model. Together, these metrics offer a comprehensive view of the LSTM Univariate model's forecasting quality on the test data.

Figure 26 highlights the performance of the LSTM Multivariate model in predicting housing data. The bluish line represents the historical standards of the 'House overall' variable, while the greenish line shows the model's predictions on the training data, demonstrating its aptitude to track past tendencies. The reddish line reflects the model's forecasts on the test data, indicating its effectiveness in forecasting new, unseen data. This comparison visually distinguishes the historical data, the model's training set predictions, and its test set predictions, helping experts evaluate the model's performance in both phases.

Figure 27 provides a summary of the LSTM Multivariate model's performance on the test dataset, highlighting essential evaluation parameters. The MSE is 0.02284, representing the average squared error in the model's predictions. The RMSE, at 0.15113, provides the average size of the errors. Additionally, the MAE is 0.123, reflecting the average absolute modification among the foreseen and real values. The MAPE of 0.184 represents the percentage performance of the model's forecasts. The MASE of 0.996 suggests a normalized measure of precision, while the R^2 score of 0.694 reveals the proportion of variance explained by the model. Together, these metrics provide a comprehensive view of the model's performance, precision, and reliability in forecasting.

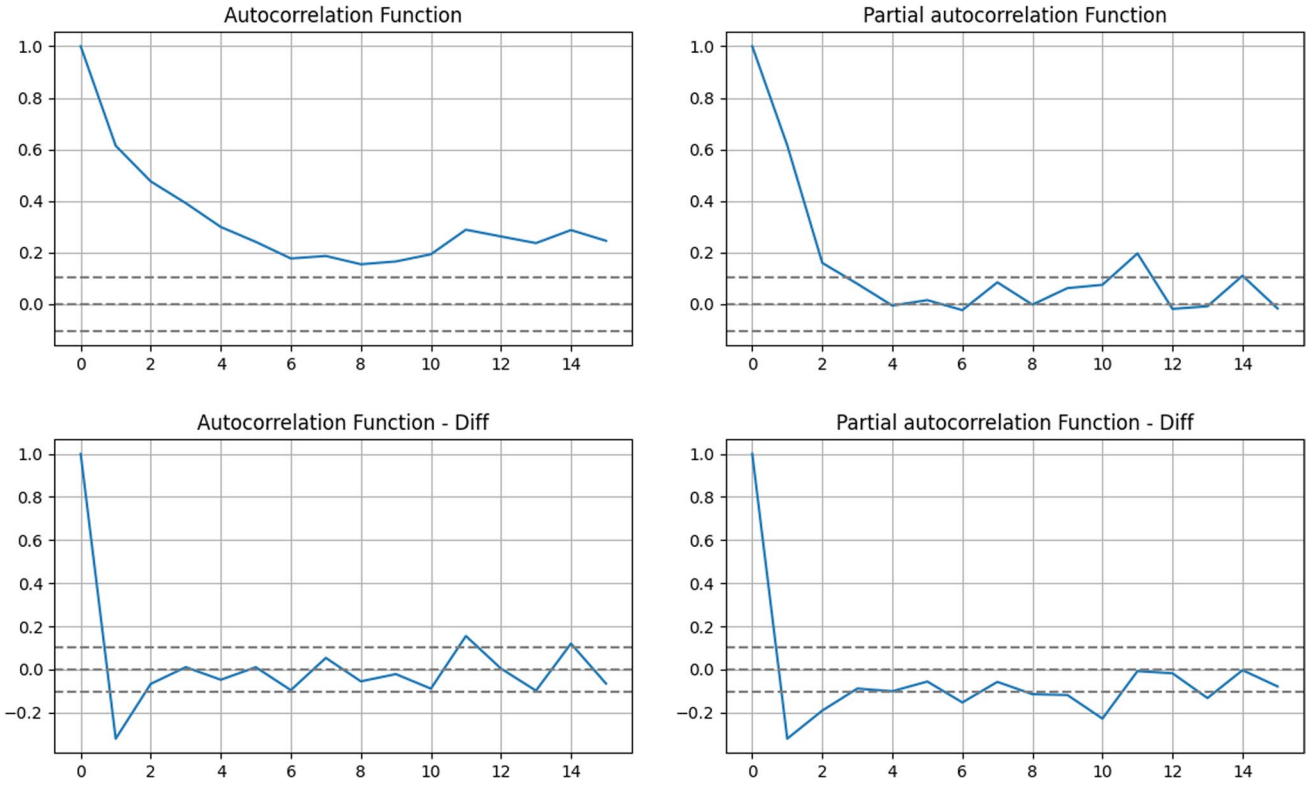


Fig. 17 Temporal analysis: ACF and PACF of daily energy consumption

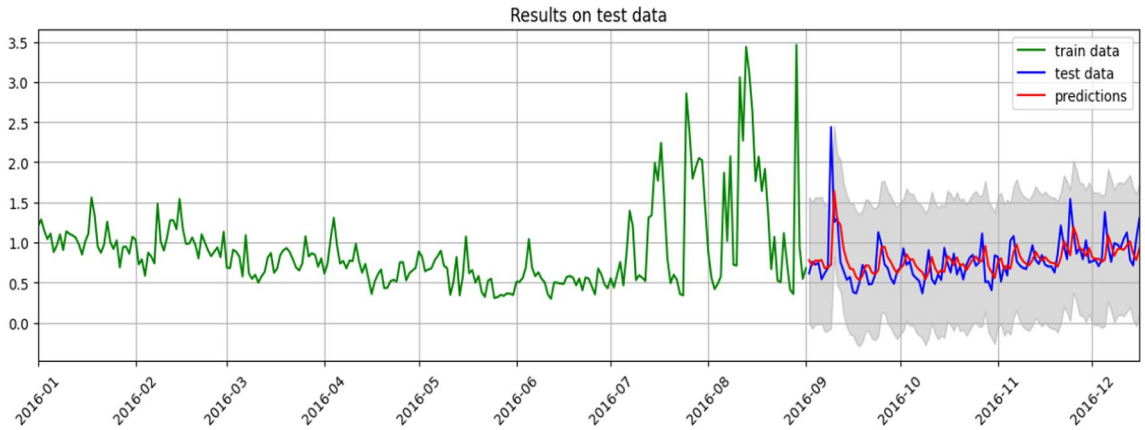


Fig. 18 Evaluation and forecasting with ARIMA model (Testing)

Figure 28 presents a comprehensive comparison of different predictive models, evaluated using several performance parameters. Separately row resembles to a precise model, counting the Baseline Model, ARIMA, Auto ARIMA, SARIMAX, LSTM Univariate, as well as LSTM Multivariate. The performance metrics shown in the columns are MSE, RMSE, MAE, MAPE, MASE, as well as R^2. Lower values for MSE, RMSE, MAE, MAPE, and MASE signify better model performance, while a higher R^2 value indicates a stronger fit to the actual data. The analysis shows that the "LSTM Multivariate" model constantly leads in performance, achieving the lowest values in most metrics and proving to be highly effective for accurate and reliable predictions.

Figure 29 displays the moving average tendency for the resampled daily 'House overall' data. The bluish line represents the real-time series, showing the overall pattern in the data. The red line shows the rolling mean, which smooths the data to highlight long-term trends and reduce short-term fluctuations. The green dashed lines indicate the upper and

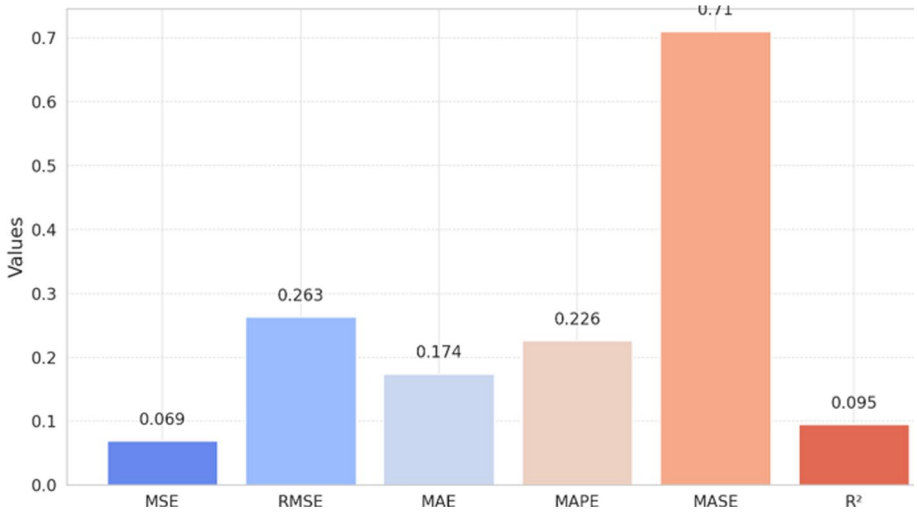


Fig. 19 Forecasting performance metrics with ARIMA model (Testing)

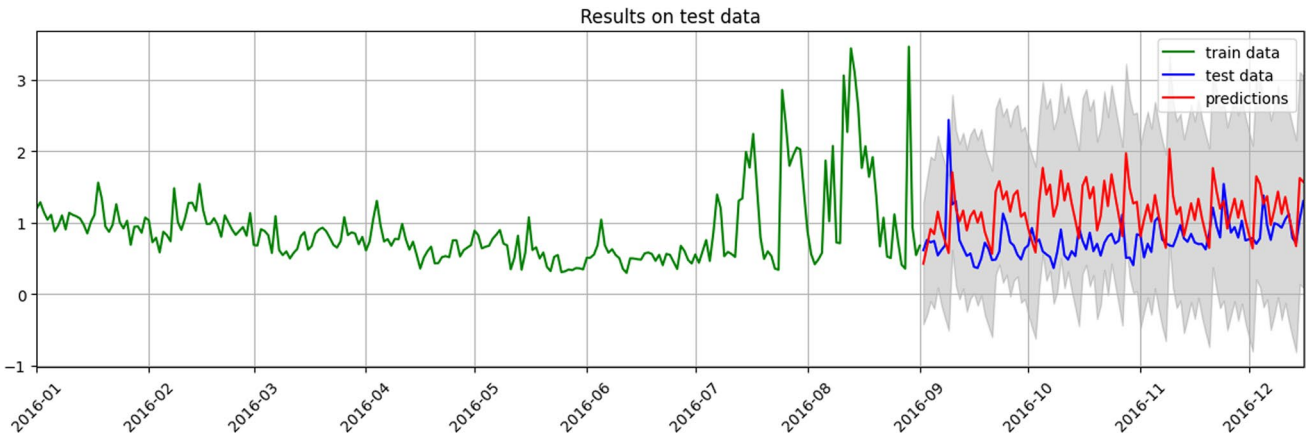


Fig. 20 Evaluation and forecasting with the auto ARIMA model on test data

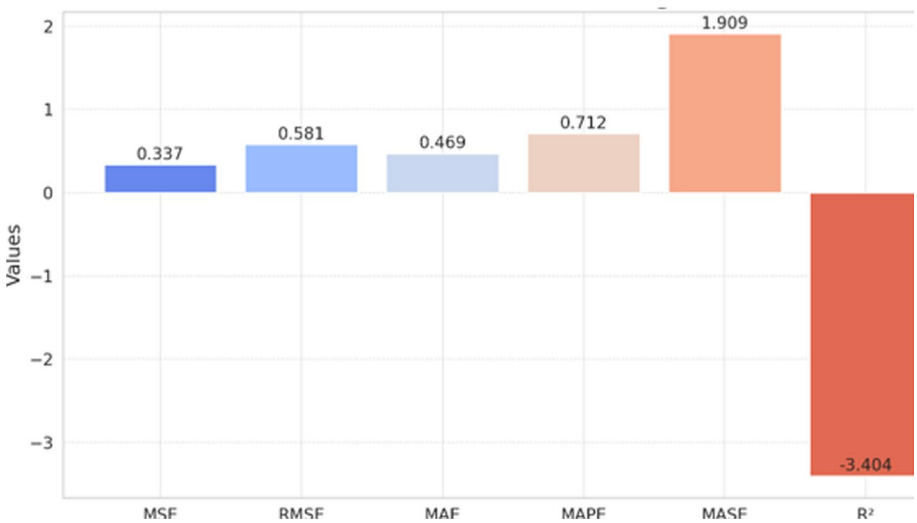


Fig. 21 Performance metrics with auto ARIMA model (Testing)

lower confidence intervals for each month, marking the typical range where most values are expected to fall. Red dots signify anomalies—points that significantly differ from the expected range of values. This graph is an effective tool for analyzing data behavior, identifying trends and anomalies, and assessing the overall reliability and pattern of the data.

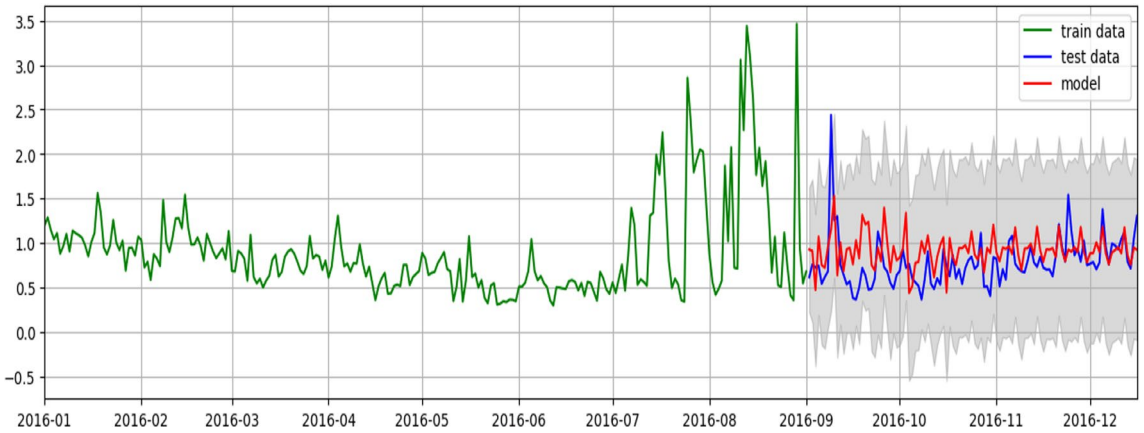


Fig. 22 Forecasting and evaluation of test data using the SARIMAX model

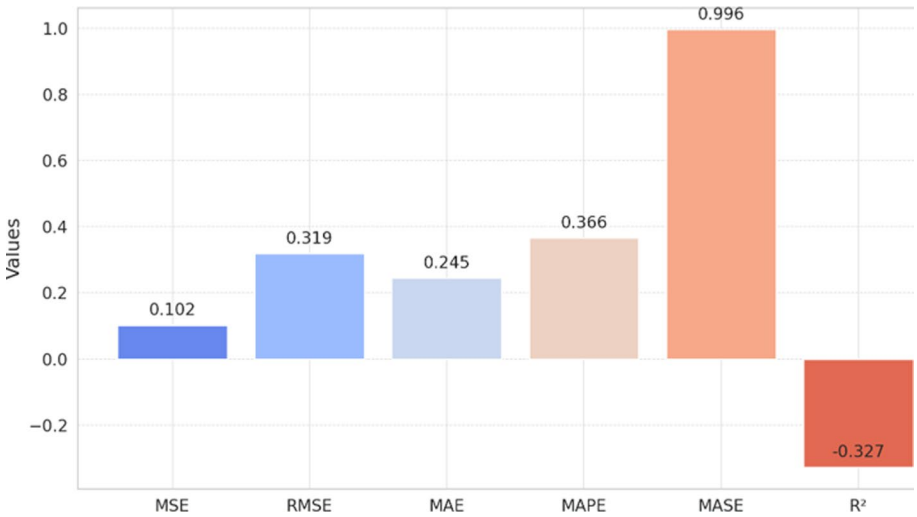


Fig. 23 Forecasting metrics using SARIMAX model (Testing)

Figure 30 delivers an anomaly detection investigation for the 'House overall' data utilizing an ARIMA model. In this graph, the red line depicts the model's predictions, while the blue line shows the actual energy usage data. The greenish dashed lines signify the upper in addition lower bounds, which are intended from the model's squared errors. Red dots signify anomalies—data points that deviate significantly from the models predicted values, falling outside these bounds. This visualization is useful for classifying rare patterns or outliers in the time series, highlighting the anomalies detected through the ARIMA model.

Figure 31 presents the results of an anomaly detection analysis for home energy consumption using an LSTM model. In this visualization, the bluish line represents the daily total energy consumption, whereas the green line shows the LSTM model's predictions derived from the training data. The shaded area between the upper and lower bounds, marked in black, illustrates the prediction uncertainty of the model. Red dots are used to denote anomalies, which are points where the actual energy consumption significantly diverges from the model's predictions. The graph, titled "LSTM Anomaly Detection," provides a clear depiction of the model's ability to recognize typical energy consumption patterns and identify unusual variations over time. The x-axis reflects time, while the y-axis corresponds to energy consumption levels.

Table 3 shows that the proposed model performance in terms of MSE is showing better performance as compared to the previously published approaches [41, 42].

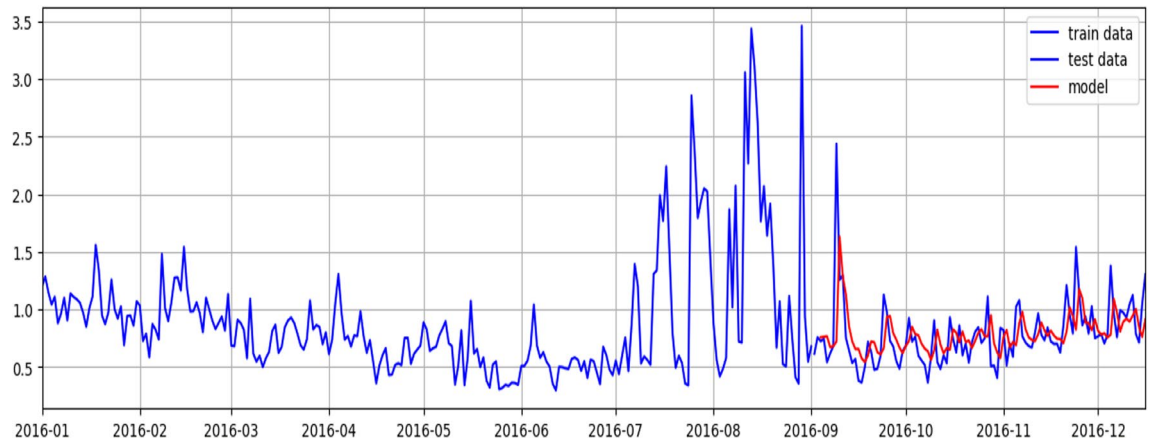


Fig. 24 Evaluation and forecasting of the LSTM univariate model on test data

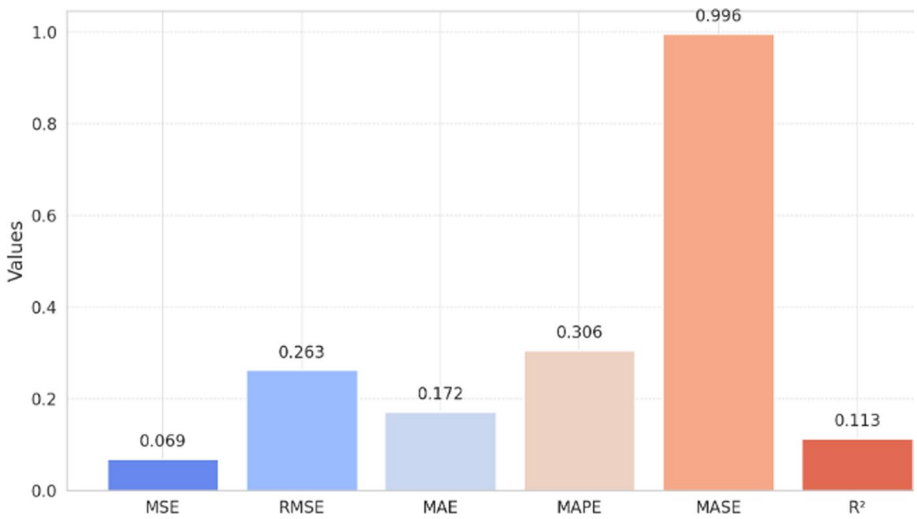


Fig. 25 Performance metrics using LSTM univariate model (Testing)

5 Conclusion

In the modern era, where energy efficiency plays a crucial role in enhancing daily living, efficiently controlling energy consumption in smart homes and buildings remains a significant challenge. Smart home technologies have advanced far beyond their initial uses, now integrating resource management, energy saving, security measures, and even health monitoring. ML techniques have revolutionized this space, allowing smart homes to learn and adapt to optimize their operations. This research explores an energy consumption model powered by ML, highlighting the LSTM Multivariate model as the most effective solution. It achieves the lowest MSE of 0.02284, indicating minimal prediction errors and high forecasting precision. The model shows impressive performance, with an RMSE of 0.15113 and strong performance in MAE and MAPE at 0.123 and 0.184, respectively. The MASE of 0.996 maintains performance relative to a simple forecast, while an R^2 value of 0.694 highlights the model's excellent fit to the data. This approves the model's effectiveness in providing accurate and reliable predictions compared to the previous approaches, paving the method for a more supportable and bright future.

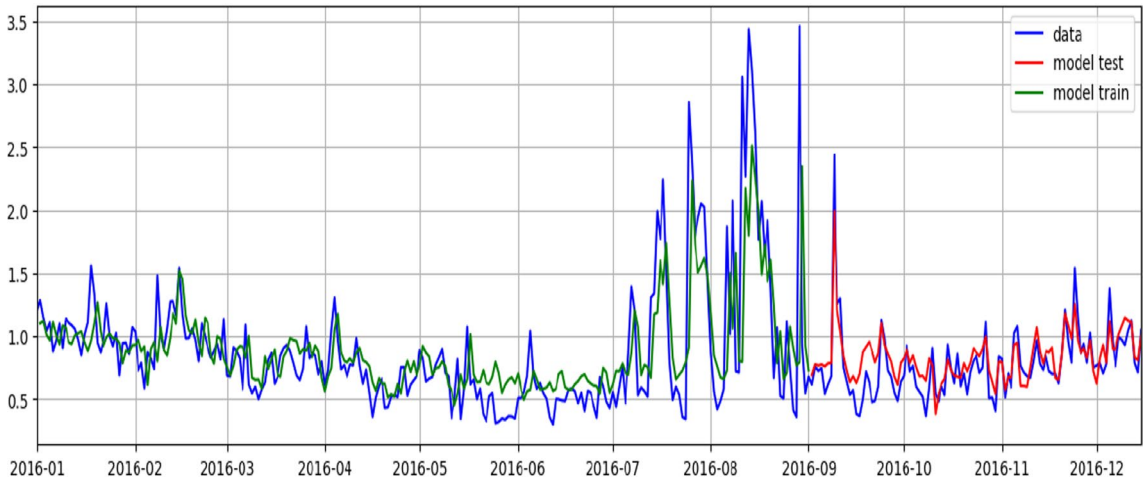
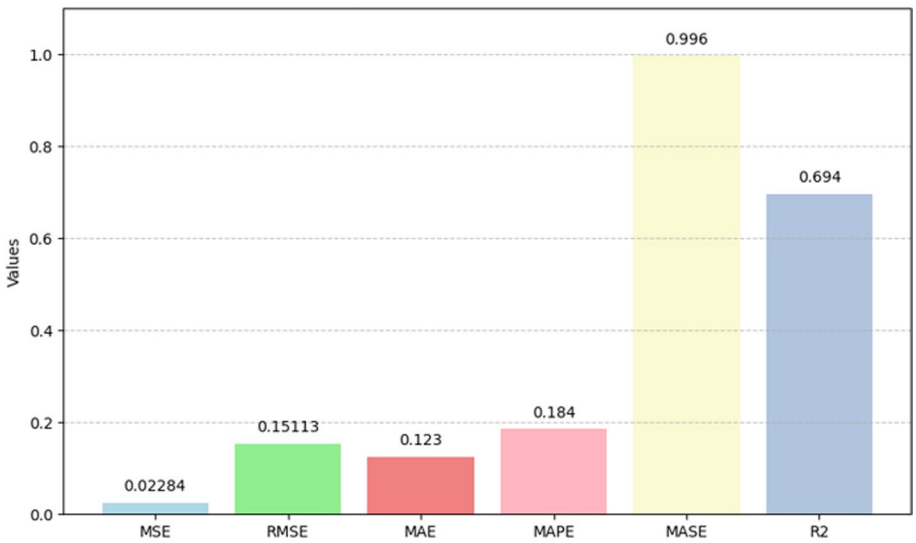


Fig. 26 Test set evaluation and forecasting using LSTM multivariate model

Fig. 27 Performance metrics LSTM multivariate model (Testing)



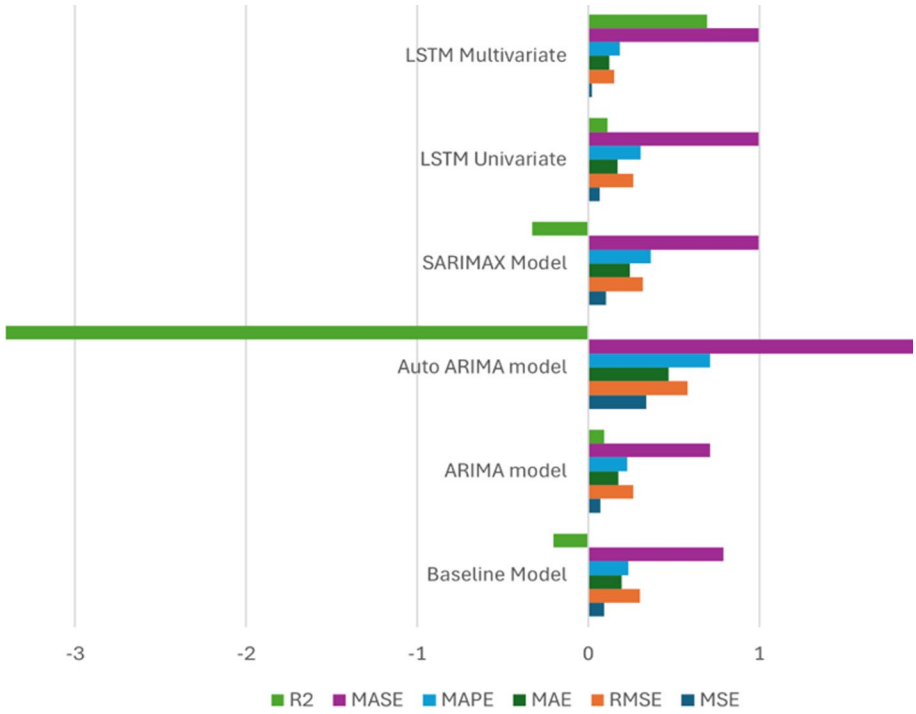


Fig. 28 Comparative analysis of model performance metrics

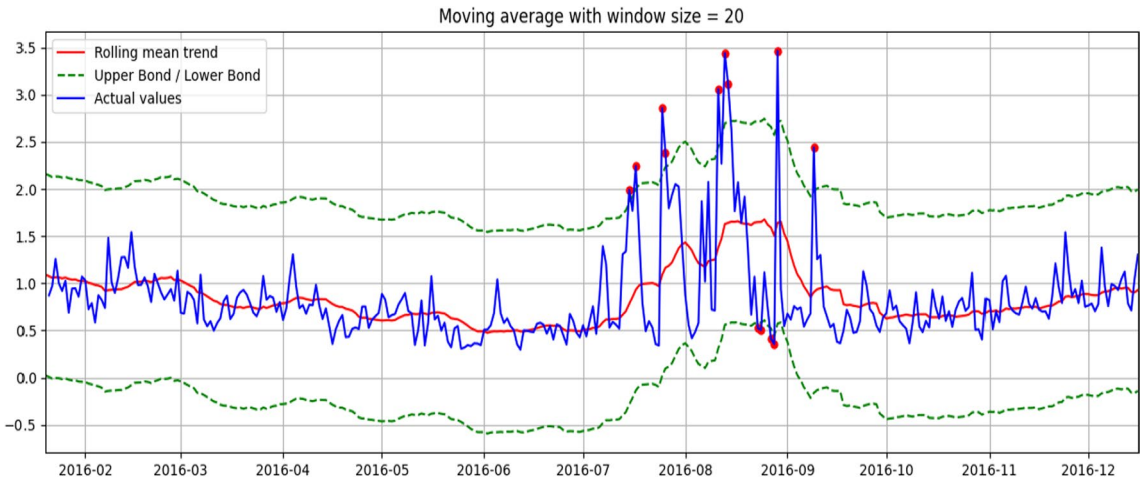


Fig. 29 Analysis of daily 'House Overall' data with moving averages and anomaly detection

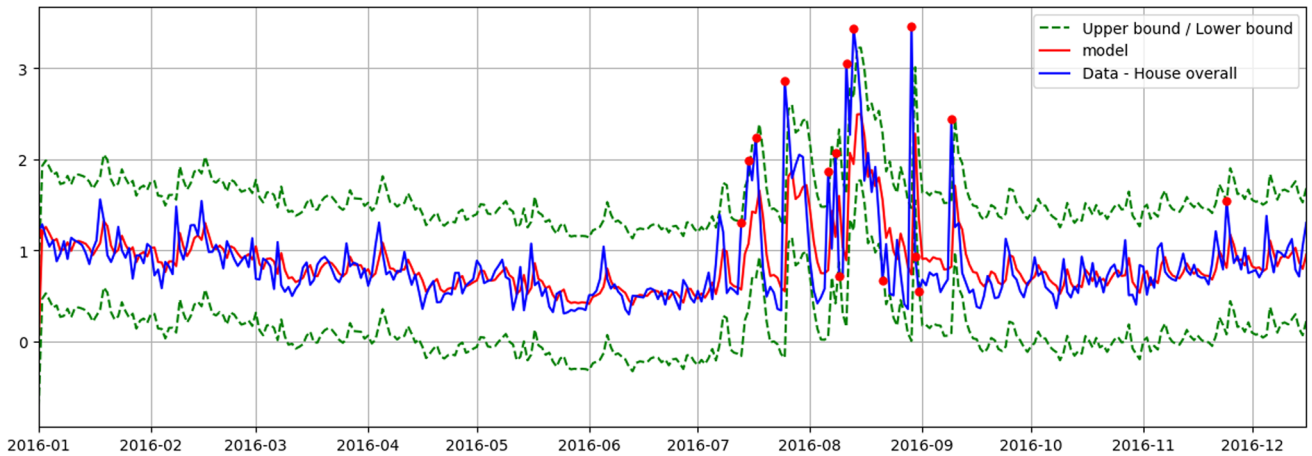


Fig. 30 ARIMA model-based anomaly detection for 'House Overall' Data

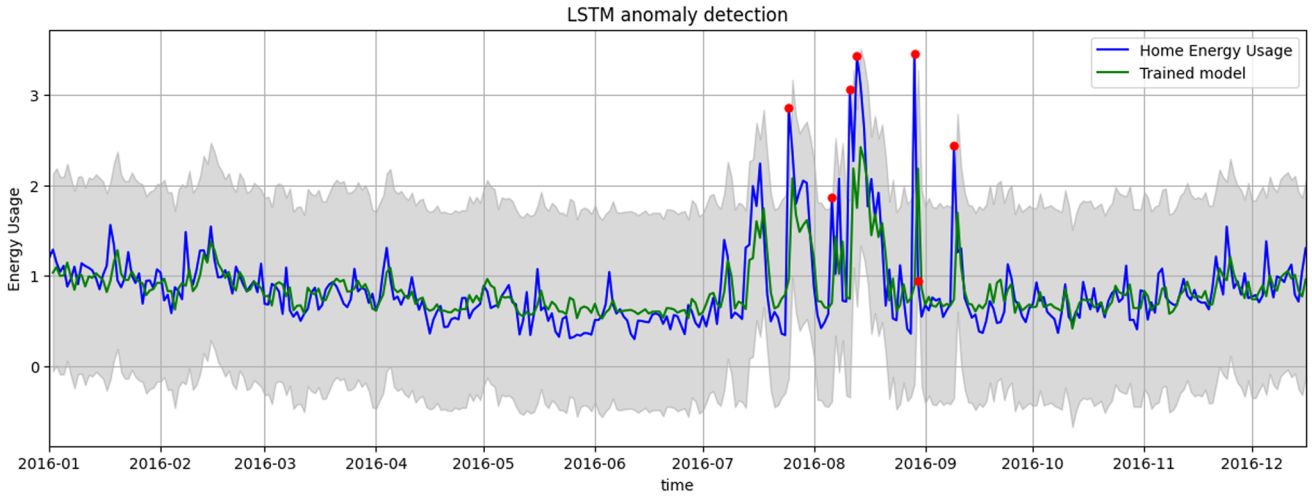


Fig. 31 LSTM model for identifying anomalies in home energy consumption

Table 3 Comparative analysis of the proposed model with previously published approaches

References	Model	MSE
Rahman et al. [41]	HW Additive	258350.1
Rahman et al. [41]	HW Multiplicative	262260.4
Rahman et al. [42]	Seasonal ARIMA	723502.2
de Mattos Neto et al. [42]	SARIMA	10
de Mattos Neto et al. [42]	AR	7
de Mattos Neto et al. [42]	MLP	5
de Mattos Neto et al. [42]	ELM	2
de Mattos Neto et al. [42]	ESN	6
de Mattos Neto et al. [42]	RBF	9
de Mattos Neto et al. [42]	Ensemble Mean	3
de Mattos Neto et al. [42]	Ensemble Median	4
de Mattos Neto et al. [42]	Ensemble MLP	8
de Mattos Neto et al. [42]	Ensemble ELM	1
Proposed model	LSTM Multivariate	0.02284

Author contributions The contributions of the authors are as follows: M.A.K, Z.S.TM, and M.S.F conceptualized the research and were responsible for writing the original draft. M.S. M.A.T.S and S.A. were responsible for the methodology and interpretation of results. Validation was carried out by Z.S.MAS M.S., and M.A.K. Re-view and editing were performed by S.A., M.S.F., and S. Abbas managed M.A. Visualization and super-vision.

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Data availability The simulation files/data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Institutional review board Not applicable.

Informed consent Not applicable.

Competing interests The authors declare no competing interests.

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